

Please check the following Questions related to basic QA

Manual Testing:

- What is the difference between QA, QC and Software Testing?
- Regression Testing & Retesting?
- What is the difference between Retesting and Regression testing?
- What are verification and validation and difference between these two?
- What is the difference between Smoke testing and Sanity Testing?
- Explain bug life cycle?
- What is the severity and priority of a bug?
- Give some examples like - High Priority and High Severity ? High Priority and Low Severity ? Low Priority and High Severity ? Low Priority and Low Severity ?
- What is the difference between build and release?
- Explain bug leakage and bug release.
- What are the key challenges of software testing?
- What is alpha and beta testing?
- What is Monkey testing?
- What is the Traceability Matrix or RTM?
- What is random testing?
- Monkey testing ? Ad-hoc testing ?
- Test plan, describe test plan
- Describe Use Case
- What is Test case, elements of test cases
- Functional & Non-Functional Testing?
- What is manual testing?
- Exploratory Testing?
- Testing principles with example
- SDLC & STLC?
- Integration testing?
- What is the testing life cycle?
- What is the difference in responsibilities of Programmers and QA analysts?
- What is the role of a Bug Tracking System?
- What methodologies do you used to develop test cases?
- Equivalence Class Partitioning & Boundary value analysis?
- Why is regression testing important?
- What is the purpose of testing?
- Details about Agile methodology
- What is end to end testing?
- Waterfall model, V model, W model, iterative model
- Positive testing, Negative testing

API Testing:

1. REST API Protocols:
 - a. GET
 - b. POST
 - c. PUT
 - d. Patch
 - e. DELETE
2. What is API and why we use API
3. Difference between Authentication and Authorization
4. Difference between Access Token and Refresh Token during authentication
5. What is collection and environment?
6. What is the base url, path parameter and query parameter?
7. Differ between header and body? Describe some header type and body types which are commonly used
8. What is pre-request script and tests?
9. What do we do in pre-request scripts and tests?
10. What is test runner?
11. What is curl?
12. What is SOAP API? Why do we use SOAP API in some cases? What is SOAP API format?
13. Difference between 4xx and 5xx?
14. Error code, response codes?

Tools:

1. Charles proxy
2. BURP suite
3. Postman
4. SOAP UI
5. AWS: Labda, EC2, Cloudwatch, DynamoDB
6. Redis
7. Docker

Coding:

1. Loops
2. Functions
3. OOP
4. Sync and Async Function
5. Basic algorithms : Binary Search, Bubble sort, Greedy algorithm, BFS, DFS, Backtracking

Network:

1. Protocols: HTTP, TCP, UDP
2. OSI model
3. How an API request works in scope of OSI model

4. Difference between HTTP & HTTPS
5. In which cases we need to use TCP & IDP. Which one is better in terms of security
6. Some scenario based questions about TCP & IDP

Automation:

1. Page Object Model
2. Element selection
3. Selectors
4. Selenium, Cypress, Playwright
5. API Automation, UI Automation

Load testing:

1. Load testing
2. Performance testing
3. Spike testing
4. Performance matrix
5. JMeter, Locust
6. Average Response Time, Error Rate, Throughput, Concurrent Users, Peak Response Time
7. Difference between Load testing & Stress testing
8. When and which cases Spike testing is needed
9. Does load testing have any specific result like Pass / Fail?
10. Basic flow of load test in JMeter

Scenario-based questions:

1. Scenario: You are testing an e-commerce website, and customers are reporting issues with the checkout process. How would you approach this problem?

- Answer: I would start by checking the checkout process step by step, looking for any UI issues, error messages, or slow loading times. I'd also review the error logs and examine the server response times. This will help identify the root cause of the issues and prioritize them for resolution.

2. Scenario: You are testing a mobile app, and it needs to work on various devices and screen sizes. How would you ensure compatibility testing?

- Answer: I would create a matrix of target devices and screen sizes. Then, I'd test the app on each combination, looking for issues related to screen layout, touch responsiveness, and device-specific functionality. Emulators and real devices can be used for testing. This ensures that the app works consistently across various platforms.

3. Scenario: You are testing a banking application, and a user reported that their account balance is not updating correctly. How would you investigate and document this issue?

- Answer: I would first try to reproduce the issue by following the user's reported steps. If the problem persists, I would document the issue by including the user's steps, expected and actual results, and any screenshots. Additionally, I'd check the server logs to identify any transaction failures or discrepancies.

4. Scenario: You are testing a software update for a complex industrial control system. How would you plan your testing strategy to ensure system stability and safety?

- Answer: I would begin with risk analysis to identify critical areas that must not fail. I'd create a detailed test plan, including unit, integration, system, and regression testing. The testing would involve simulating real-world scenarios and ensuring safety mechanisms are in place, like emergency stop functionality.

5. Scenario: You are working on a tight deadline for a software release, and you have a large number of test cases to execute. What strategies would you employ to meet the deadline without compromising quality?

- Answer: To meet the deadline, I would prioritize test cases based on their criticality, focus on high-impact areas, and automate repetitive tests. I'd also collaborate closely with the development team to ensure quick defect resolution. Continuous communication and risk-based testing would be key to balancing speed and quality.

6. Scenario: You are testing a healthcare application, and you come across a potential HIPAA compliance issue. What steps would you take to address this concern?

- Answer: I would immediately report the compliance concern to the project manager, security team, and legal department. I would stop testing any areas related to sensitive patient data until the compliance issue is addressed. It's crucial to follow legal and ethical guidelines when handling such issues.

7. Scenario: You are testing a GPS navigation app, and users have reported inaccuracies in location tracking. How would you approach this issue?

- Answer: I'd verify the issue by using the app on different devices and locations. I'd also cross-reference the app's data with other trusted mapping services to identify discrepancies. Additionally, I'd check the app's settings and permissions to ensure correct usage.

8. Scenario: You are tasked with testing a new feature in a web application that requires interaction with third-party APIs. How would you ensure the reliability of this feature?

- Answer: I'd thoroughly test the feature's interaction with the third-party APIs, covering various scenarios such as successful responses, timeouts, and error handling. I'd also monitor the API responses for consistency and performance. Integration testing with these APIs is crucial to ensure reliable functionality.

9. Scenario: You are testing a video streaming app, and users have reported buffering issues. How would you troubleshoot and identify the root cause of this problem?

- Answer: I'd start by checking the user's network connection and device performance. If those are not the causes, I'd review server logs to identify any latency issues or server-side problems.

I'd also check the app's caching and buffering mechanisms to ensure they are functioning optimally.

10. Scenario: You are testing a software application for a global audience. How would you handle localization and internationalization testing?

- Answer: I'd begin by creating test cases for language support, date and time formats, currency symbols, and regional settings. I'd also test the application's ability to adapt to different languages, character sets, and cultures. Emulators, simulators, or real devices representing different regions would be used for testing.

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16. Scenario: You are testing a social media platform, and users have reported that posts are disappearing after they're published. How would you investigate and address this issue?

- Answer: I would begin by creating test cases to replicate the problem. I'd check the database to ensure data persistence and review the post publishing process, looking for any errors or timing issues. Consistency in post visibility would be a priority.

17. Scenario: You are testing a mobile banking app, and you receive a complaint that the app crashes when users attempt to transfer money. How would you troubleshoot and document this issue?

- Answer: First, I would attempt to reproduce the issue on multiple devices. If successful, I'd document the crash with detailed steps, device information, and screenshots. I'd also check the error logs to identify any exceptions or crashes. Communication with the development team is key to resolving this critical issue.

18. Scenario: You are testing a new feature in a web application that involves user authentication and authorization. How would you ensure the security and reliability of this feature?

- Answer: I'd create test cases to cover various scenarios, including valid and invalid login attempts, role-based access, and data security. I'd also conduct penetration testing to identify vulnerabilities. Collaboration with the security team and compliance with best practices are crucial.

19. Scenario: You are testing a gaming application, and players have reported graphical glitches and rendering issues. How would you approach identifying and addressing these visual problems?

- Answer: I'd test the game on a variety of devices with different screen resolutions, GPUs, and hardware specifications. Screenshots and screen recordings would be used to document glitches. Investigating rendering code, graphics assets, and OpenGL settings may be necessary to resolve visual issues.

20. Scenario: You are testing a file-sharing application, and users report that uploaded files are occasionally corrupted. How would you investigate and address this data integrity issue?

- Answer: I would start by examining the upload process and the handling of different file types. Testing various file formats, sizes, and network conditions is essential. I'd also check for error handling during the upload process and implement checksum or hash verification to ensure data integrity.